

20/06/2026

Assignment No- 03

Week - 3

1) Student Marks Analysis

```
→ students = [
    ["Amit", 78, 85, 92],
    ["Sara", 65, 72, 68],
    ["John", 90, 88, 95],
    ["Fatima", 85, 52, 48],
    ["Ali", 80, 79, 81]
]

print("Students Results:")
topper = ""
highest_percentage = 0

for student in students:
    name = student[0]
    total = sum(student[1:])
    percentage = total / 3
    print(f"{name}: Total= {total}, Percentage= {percentage: .2f}%")

    if percentage > highest_percentage:
        highest_percentage = percentage
        topper = name

print("\nTopper: ", topper)
print("Students below 50%:")
```



```
for student in students:
```

```
    percentage = sum(student[1:])/3
```

```
    if percentage < 50:
```

```
        print(student[0])
```

2) Store inventory

```
→ p = [{"Laptop", 25}, {"Mouse", 150},
        {"Keyboard", 80}, {"Monitor", 18},
        {"printer", 5}]
```

```
print("Stock < 20: ")
```

```
for i, j in p:
```

```
    if j < 20: print(i, j)
```

```
print("Total = ", sum(j for i, j in p))
```

```
print("Stored = ", sorted(p, key = lambda
                           x: x[1]))
```

```
import matplotlib.pyplot as plt
```

```
plt.bar([i for i, j in p], [j for i, j
                           in p])
```

```
plt.show()
```


3) Employee Salaries -

```
→ e = [{"HR", 25000}, {"IT", 45000},
        {"IT", 55000}, {"Sales", 30000},
        {"HR", 27000}]
```

```
for d in ["HR", "IT", "Sales"]:
    print(d, "Avg = ", sum(s for x, s
                           in e if x == d) / len
          (x for x, s in e if x == d))
```

```
print(Highest = ", max(e, key=lambda
      x: x[1]))
```

```
import matplotlib.pyplot as plt
plt.bar(["HR", "IT", "Sales"],
        [26000, 50000, 29000])
plt.show()
```

4) Missing Values challenge.

```
→ import pandas as pd
```

```
d = {"Name": ["A", "B", "C", "D", "E"],
     "Age": [2, None, 23, None, 25],
     "Marks": [78, 85, None, 90, 88]}
```

```
df = pd.DataFrame(d)
```

```
print(df.isnull().sum())
```



```
df.fillna(df.mean(numeric_only=True),
          inplace=True)
```

```
print(df)
```

5) Monthly sales -

```
→ m = ["Jan", "Feb", "Mar", "Apr", "May",
        "Jun"]
```

```
s = [25000, 30000, 28000, 40000, 42000,
      38000]
```

```
print("Avg = ", sum(s)/len(s))
```

```
print("Highest = ", m[s.index(max(s))])
```

```
print("Lowest = ", m[s.index(min(s))])
```

```
import matplotlib.pyplot as plt
```

```
plt.plot(m, s, marker="o")
```

```
plt.show()
```

6) Online orders -

```
→ o = [[101, "Mobile", 2, 15000],
        [102, "Laptop", 1, 55000],
        [103, "Headphones", 8, 2000],
        [104, "Tablet", 2, 18000]]
```

```
v = [i[2]*i[3] for i in o]
```


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```

for i, j in zip(o, v):
    print(i[0], j)

print("Highest = ", o[v.index(max(v))][0])
print("Revenue = ", sum(v))

```

7) Duplicate Records

```

→ s = [{"Amit", 22}, {"Sara", 21},
        {"Amit", 22}, {"John", 23},
        {"Sara", 21}]

```

```

u = []

```

```

for i in s:
    if i not in u:
        u.append(i)
    else:
        print("Duplicate = ", i)

print("Final = ", u)

```

8) Temperature Monitoring

```

→ t = [32, 34, 35, 33, 38, 40, 42, 39, 37, 36]

print("Avg = ", sum(t) / len(t))
print("Hottest Day = ", t.index(max(t)) + 1)

```



```
print("coldest Day = ", t.index(min(t))+1)
```

```
import matplotlib.pyplot as plt
```

```
plt.plot(t, markers="o")
```

```
plt.show()
```

9) Exam Performance

```
→ import numpy as np
import matplotlib.pyplot as plt
```

```
m = [78, 85, 92, 65, 80]
```

```
s = [82, 88, 96, 76, 75]
```

```
print("Math Avg = ", sum(m)/len(m))
```

```
print("Sci Avg = ", sum(s)/len(s))
```

```
print("Correlation = ", np.corrcoef(m,s)
      [0,1])
```

```
plt.scatter(m,s)
```

```
plt.xlabel("maths")
```

```
plt.ylabel("science")
```

```
plt.show()
```